Applied Linear Algebra in Data Analysis

Orthogonality

Sivakumar Balasubramanian

Department of Bioengineering
Christian Medical College, Bagayam
Vellore 632002

Some more Vector Space Concepts: Inner Product

- **Standard inner product** is defined as the following,

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i, \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

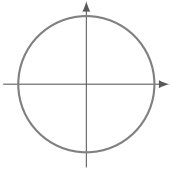
  For complex vectors: $\mathbf{x}^* \mathbf{y} = \sum_{i=1}^{n} \overline{x}_i y_i, \quad \mathbf{x}, \mathbf{y} \in \mathbb{C}^n$

- **Properties**
  - $\mathbf{x}^\top \mathbf{x} > 0, \quad \forall \mathbf{x} \neq 0$ and $\mathbf{x}^\top \mathbf{x} = 0 \Leftrightarrow \mathbf{x} = 0$
  - *Commutative*: $\mathbf{x}^\top \mathbf{y} = \mathbf{y}^\top \mathbf{x}$
  - *Associativity with scalar multiplication*: $(\alpha \mathbf{x})^\top \mathbf{y} = \alpha \left(\mathbf{x}^\top \mathbf{y}\right)$
  - *Distributivity with vector addition*: $(\mathbf{x} + \mathbf{y})^\top \mathbf{z} = \mathbf{x}^\top \mathbf{z} + \mathbf{y}^\top \mathbf{z}$

Some more Vector Space Concepts: Norms

- Norm is a measure of the size of a vector.
- *Euclidean norm* of a n-vector $\mathbf{x} \in \mathbb{R}^n$ is defined as, $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}} = \sqrt{\sum_{i=1}^{n} x_i^2}$.
- $\|\mathbf{x}\|_2$ is a measure of the length of the vector $\mathbf{x}$.
- Any function of the form $\|\bullet\| : \mathbb{R}^n \longrightarrow \mathbb{R}_{\geq 0}$ is a valid norm, provided it satisfies the following properties.
  - *Definiteness.* $\|\mathbf{x}\| = 0 \iff x = 0$
  - *Non-negativity.* $\|\mathbf{x}\| \geq 0$
  - *Non-negative homogeneity.* $\|\beta \mathbf{x}\| = |\beta| \, \|\mathbf{x}\| \, , \, \beta \in \mathbb{R}$
  - *Triangle inequality.* $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$

Some more Vector Space Concepts: Geometry of Norms



$$\|\mathbf{x}\|_2 = \left(\sum_{i=1}^{n} x_i^2\right)^{\frac{1}{2}}$$

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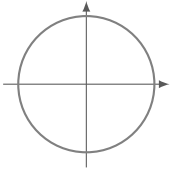

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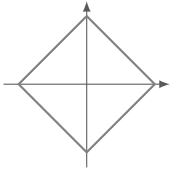

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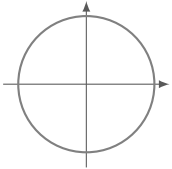

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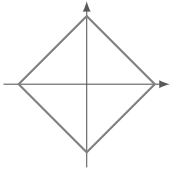

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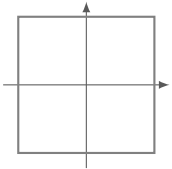

$$\|\mathbf{x}\|_\infty = \max_{1 \le i \le n} |x_i|$$

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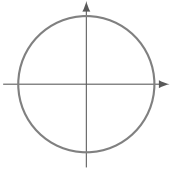

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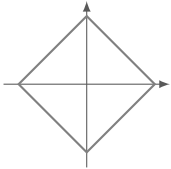

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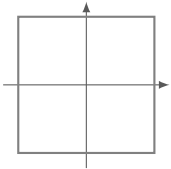

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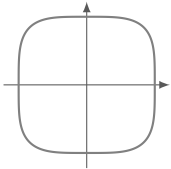
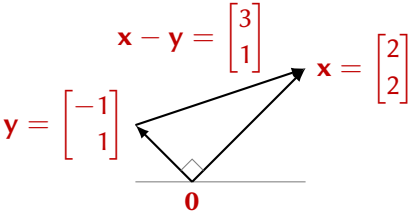


$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} x_i^p\right)^{\frac{1}{p}}$$

Some more Vector Space Concepts: Orthogonality

- Orthogonality is the idea of two vectors being perpendicular, $\mathbf{x} \perp \mathbf{y}$.



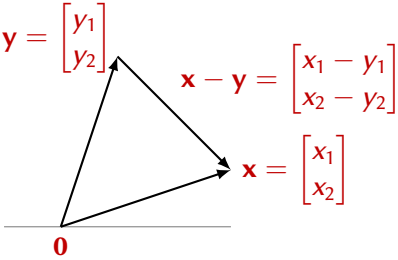
Using the Pythagonean theorem, $\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \implies \mathbf{x}^\top \mathbf{y} = 0$$

- We extend this to the n-dimensional case and define two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ being orthogonal, if

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i = 0$$

Some more Vector Space Concepts: Angle between vectors



- Inner products are used for projecting a vector onto another vector or a subspace.
- It is also a measure of similarity between two vectors, $\cos(\theta) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$
- **Cauchy-Bunyakovski-Schwartz Inequality**:

$$\left| \mathbf{x}^\top \mathbf{y} \right| \leq \|\mathbf{x}\| \, \|\mathbf{y}\| \, , \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Some more Vector Space Concepts: Basis

- A linearly independent set of n-vectors from $\mathbb{R}^n$, of size n, is called a *basis* for $\mathbb{R}^n$.

- Any n-vector from $\mathbb{R}^n$ can be represented as a *unique* linear combination of the elements of the basis.

- Consider the basis $\{\mathbf{x}_i\}_{i=1}^n$, $\mathbf{x}_i \in \mathbb{R}^n$. Any vector $\mathbf{y} \in \mathbb{R}^n$ can be represented as a linear combination of $\mathbf{x}_i$s, $\mathbf{y} = \sum_{i=1}^n \alpha_i \mathbf{x}_i$. This is called the *expansion of* $\mathbf{y}$ in the $\{\mathbf{x}_i\}_{i=1}^n$ basis.

- The numbers α_i are called the *coefficients* of the expansion of $\mathbf{y}$ in the $\{\mathbf{x}_i\}_{i=1}^n$ basis.

Some more Vector Space Concepts: Basis

- **Orthogonal vectors**: A set of vectors $\{\mathbf{x}_i\}_{i=1}^n$ is *(mutually) orthogonal* if $\mathbf{x}_i \perp \mathbf{x}_j$ for all $i, j \in \{1, 2, 3, \ldots n\}$ and $i \neq j$.
- This set is called **orthonormal** if its elements are all of unit length $\|\mathbf{x}_i\|_2 = 1$ for all $i \in \{1, 2, 3, \ldots n\}$.

$$\mathbf{x}_i^\top \mathbf{x}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

- An orthonormal collection of vectors is linearly independent.
- Consider an orthonormal basis $\{\mathbf{x}_i\}_{i=1}^n$. The expansion of a vector $\mathbf{y}$ is given by,

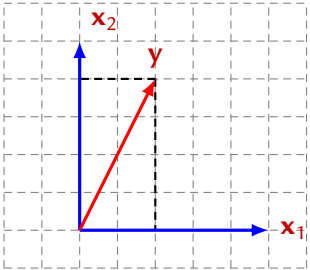
$$\mathbf{y} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \alpha_3 \mathbf{x}_3 + \cdots + \alpha_n \mathbf{x}_n$$

$$\mathbf{x}_i^\top \mathbf{y} = \alpha_1 \mathbf{x}_i^\top \mathbf{x}_1 + \alpha_2 \mathbf{x}_i^\top \mathbf{x}_2 + \alpha_3 \mathbf{x}_i^\top \mathbf{x}_3 + \cdots + \alpha_n \mathbf{x}_i^\top \mathbf{x}_n = \alpha_i$$

Representing a Vector in an Orthonormal Basis

- Thus, we can rewrite this as,

$$\mathbf{y} = \left(\mathbf{y}^\top \mathbf{x}_1\right)\mathbf{x}_1 + \left(\mathbf{y}^\top \mathbf{x}_2\right)\mathbf{x}_2 + \left(\mathbf{y}^\top \mathbf{x}_3\right)\mathbf{x}_3 + \cdots + \left(\mathbf{y}^\top \mathbf{x}_n\right)\mathbf{x}_n$$



Some more Vector Space Concepts: Addition of Subspaces

Consider two subspaces $\mathcal{V}, \mathcal{W} \subset \mathbb{R}^n$. We can define the addition of two subspaces as,

Definition

The sum of two subspaces $\mathcal{V}$ and $\mathcal{W}$ is the set of all vectors that can be written as the sum of a vector from $\mathcal{V}$ and a vector from $\mathcal{W}$.

$$\mathcal{U} = \mathcal{V} + \mathcal{W} = \{\mathbf{v} + \mathbf{w} \mid \forall \mathbf{v} \in \mathcal{V}, \mathbf{w} \in \mathcal{W}\} \subset \mathbb{R}^n$$

The resulting set will be a subspace of $\mathbb{R}^n$.

Why is $\mathcal{U}$ a subspace of $\mathbb{R}^n$?

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Why is $\mathcal{U}$ a subspace of $\mathbb{R}^n$? It is closed under vector scaling and addition.

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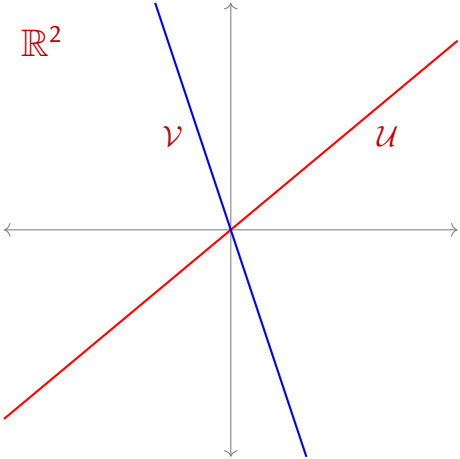

What is the sum of the two subspaces $\mathcal{U}$ and $\mathcal{V}$?

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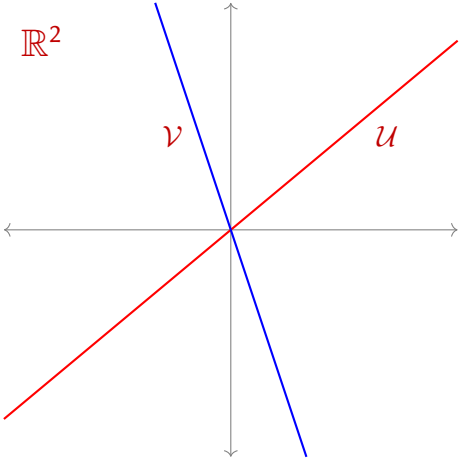

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The sum of the two subspaces is $\mathbb{R}^2$.

$$\mathcal{U} + \mathcal{V} = \mathbb{R}^2$$

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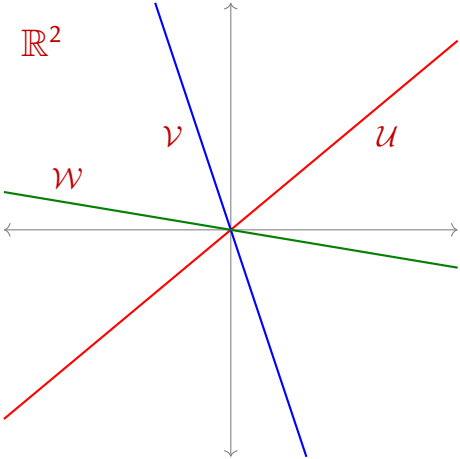

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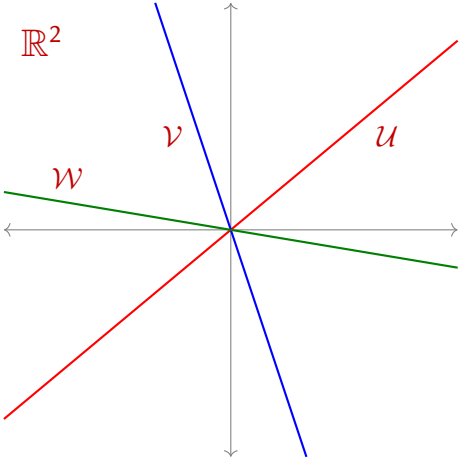

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Let's look at subspaces from $\mathbb{R}^3$.

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What is the difference between examples 3 and 4?

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What is the difference between examples 3 and 4? In example 3, $\mathcal{U} \cap \mathcal{V} = \{\mathbf{0}\}$, while in example 4, $\mathcal{U} \cap \mathcal{V} \neq \{\mathbf{0}\}$.

Some more Vector Space Concepts: Completmentary Subspaces

Consider two subspaces $\mathcal{V}, \mathcal{W} \subset \mathbb{R}^n$. We can define the addition of two subspaces as,

Definition

Two subspaces $\mathcal{V}$ and $\mathcal{W}$ are complementary if $\mathcal{V} + \mathcal{W} = \mathbb{R}^n$ and $\mathcal{V} \cap \mathcal{W} = \{\mathbf{0}\}$.

Complementary subspaces do not share anything in common (except for the zero vector) and together they span the entire space. This means there is no redundancy between the two subspaces.

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Which of the following pairs of subspaces are complementary in $\mathbb{R}^2$?

- $\mathcal{V} = \text{span}\left(\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}\right)$ and $\mathcal{W} = \text{span}\left(\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}\right)$

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- $\mathcal{V} = \text{span}\left(\left\{\begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}\right)$ and $\mathcal{W} = \text{span}\left(\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}\right)$

Some more Vector Space Concepts: Completmentary Subspaces

Consider two subspaces $\mathcal{V}, \mathcal{W} \subset \mathbb{R}^n$. We can define the addition of two subspaces as,

Definition

Two subspaces $\mathcal{V}$ and $\mathcal{W}$ are complementary if $\mathcal{V} + \mathcal{W} = \mathbb{R}^n$ and $\mathcal{V} \cap \mathcal{W} = \{\mathbf{0}\}$.

Complementary subspaces do not share anything in common (except for the zero vector) and together they span the entire space. This means there is no redundancy between the two subspaces.

Which of the following pairs of subspaces are complementary in $\mathbb{R}^2$?

- $\mathcal{V} = \text{span}\left(\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}\right)$ and $\mathcal{W} = \text{span}\left(\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}\right)$ Yes.
- $\mathcal{V} = \text{span}\left(\left\{\begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}\right)$ and $\mathcal{W} = \text{span}\left(\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}\right)$ No.

Some more Vector Space Concepts: Completmentary Subspaces

The idea of complementary subspace can be extended to more than two subspaces.

Consider the set of subspace $\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3, \ldots, \mathcal{V}_k \subset \mathbb{R}^n$. We say that these subspaces form a direct sum decomposition of $\mathbb{R}^n$, if the following two conditions are satisfied,

- $\mathcal{V}_1 + \mathcal{V}_2 + \mathcal{V}_3 + \cdots + \mathcal{V}_k = \mathbb{R}^n$
- $\mathcal{V}_i \cap \sum_{j=1, j\neq i}^{k} \mathcal{V}_j = \{\mathbf{0}\}$

Notice that this has the flavor of linear independence of a set of vectors.

Some more Vector Space Concepts: Components from a Subspace

Consider the set of subspace $\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3, \ldots, \mathcal{V}_k \subset \mathbb{R}^n$.

We can write any vector $\mathbf{x} \in \mathbb{R}^n$ as the sum of vectors from each of the k subspaces,

$$\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3 + \cdots + \mathbf{x}_k, \quad \mathbf{x}_i \in \mathcal{V}_i$$

where, $\mathbf{x}_i \in \mathcal{V}_i$ is called the component of $\mathbf{x}$ from the subspace $\mathcal{V}_i$.

When the subspaces form a direct sum decomposition of $\mathbb{R}^n$, then the components $\mathbf{x}_i$ are unique.

Else, there are infinitely many multiple ways to write $\mathbf{x}$ as the sum of vectors from each of the k subspaces.

Can you now notice the similarity to the idea of linear independence of a set of vectors?

Some more Vector Space Concepts: Orthogonal Completments

Just like the idea of orthogonal vectors, we can define orthogonal subspaces.

Definition

Two subspaces $\mathcal{V}$ and $\mathcal{W}$ are orthogonal if every vector in $\mathcal{V}$ is orthogonal to every vector in $\mathcal{W}$.

$$\mathbf{v}^\top \mathbf{w} = 0, \quad \forall \mathbf{v} \in \mathcal{V}, \mathbf{w} \in \mathcal{W} \implies \mathcal{V} \perp \mathcal{W}$$

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Which of the following pairs of subspaces are orthogonal in $\mathbb{R}^2$?

- $\mathcal{V} = \text{span}\left(\left\{\begin{bmatrix}1\\2\end{bmatrix}\right\}\right)$ and $\mathcal{W} = \text{span}\left(\left\{\begin{bmatrix}2\\-1\end{bmatrix}\right\}\right)$

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- $\mathcal{V} = \text{span}\left(\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}\right)$ and $\mathcal{W} = \text{span}\left(\left\{\begin{bmatrix}0\\0\end{bmatrix}\right\}\right)$ Yes. Every subspace is orthogonal to the zero subspace.

Some more Vector Space Concepts: Orthogonal Completments

Definition

Two subspaces $\mathcal{V}, \mathcal{W} \in \mathbb{R}^n$ are **orthogonal complements** of $\mathbb{R}^n$ if they are orthogonal and their sum spans $\mathbb{R}^n$.

$$\mathcal{V} \perp \mathcal{W} \quad \text{and} \quad \mathcal{V} + \mathcal{W} = \mathbb{R}^n \implies \mathcal{V}^\perp = \mathcal{W} \quad \text{and} \quad \mathcal{W}^\perp = \mathcal{V}$$

State if the orthogonal complements in $\mathbb{R}^3$ for the following subspaces.

- $\mathcal{V} = \text{span}\left(\{\mathbf{e}_1, \mathbf{e}_3\}\right)$.

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- $\mathcal{V} = \text{span}\left(\{\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3\}\right)$.

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- $\mathcal{V} = \text{span}\left(\{\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3\}\right)$. $\mathcal{V}^\perp$ set of all vectors orthogonal to $\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$.
- $\mathcal{V} = \text{span}\left(\{\mathbf{0}\}\right)$.

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- $\mathcal{V} = \text{span}\left(\{\mathbf{0}\}\right)$. $\mathcal{V}^\perp = \mathbb{R}^3$.

Relationship between the Four Fundamental Subspaces of A

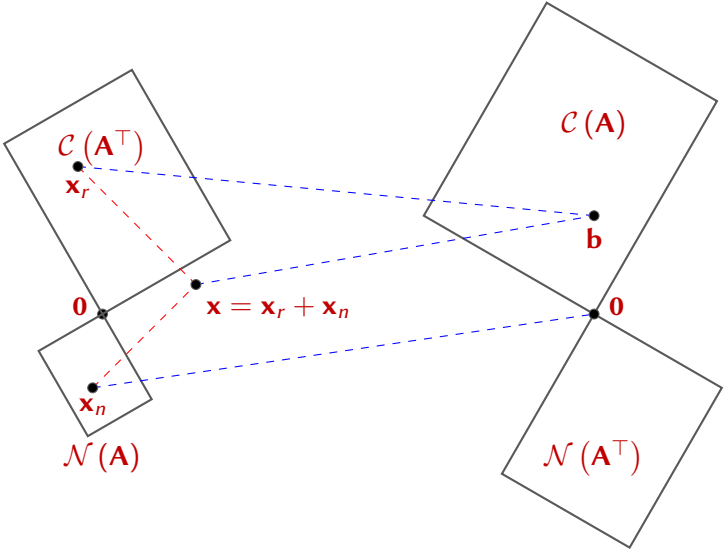
- $\mathcal{C}(\mathbf{A}), \mathcal{N}(\mathbf{A}^\top) \subseteq \mathbb{R}^n$ are orthogonal complements.

$$\mathcal{C}(\mathbf{A}) \perp \mathcal{N}\left(\mathbf{A}^\top\right) \implies \mathcal{C}(\mathbf{A}) + \mathcal{N}\left(\mathbf{A}^\top\right) = \mathbb{R}^n$$

- $\mathcal{C}(\mathbf{A}^\top), \mathcal{N}(\mathbf{A}) \subseteq \mathbb{R}^m$ are orthogonal complements.

$$\mathcal{C}\left(\mathbf{A}^\top\right) \perp \mathcal{N}(\mathbf{A}) \implies \mathcal{C}\left(\mathbf{A}^\top\right) + \mathcal{N}(\mathbf{A}) = \mathbb{R}^m$$

Relationship between the Four Fundamental Spaces



Relationship between the Four Fundamental Spaces

- $\mathbf{x}_r$ and $\mathbf{x}_n$ are the components of $\mathbf{x} \in \mathbb{R}^m$ in the row space and nullspace of $\mathbf{A}$.
- **Nullspace** $\mathcal{N}(\mathbf{A})$ is mapped to $\mathbf{0}$.

$$\mathbf{A}\mathbf{x}_n = \mathbf{0}$$

- **Row space** $\mathcal{C}\left(\mathbf{A}^\top\right)$ is mapped to the **column space** $\mathcal{C}(\mathbf{A})$.

$$\mathbf{A}\mathbf{x}_r = \mathbf{A}\left(\mathbf{x}_r + \mathbf{x}_n\right) = \mathbf{A}\mathbf{x} = \mathbf{b}$$

- The mapping from the **row space** to the **column space** is invertible, i.e. every $\mathbf{x}_r$ is mapped to a unique element in $\mathcal{C}(\mathbf{A})$

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- The mapping from the **row space** to the **column space** is invertible, i.e. every $\mathbf{x}_r$ is mapped to a unique element in $\mathcal{C}(\mathbf{A})$
- What sort of mapping does $\mathbf{A}^\top$ do?

Orthogonal Projection onto Subspaces

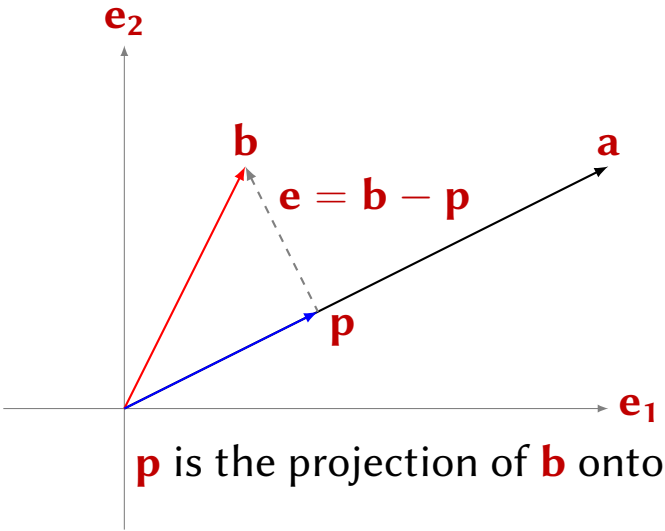
Consider $\mathbf{Ax} = \mathbf{b}$. $\mathbf{x}$ is an arbitrary solution to this equation.

How do we find the row-space component and the nullspace component of this vector?

This brings us to the idea of "orthogonal" projection onto a subspace. $\Longrightarrow$ There are "oblique" projections as well!

This a simple, but powerful idea. You are likely very familiar with this process at least geometrically from your vector algebra and plotting points on a graph sheet.

Orthogonal Projection onto Subspaces



$\mathbf{p}$ is the projection of $\mathbf{b}$ onto $\mathbf{a}$.

Orthogonal Projection onto Subspaces

$\|\mathbf{e}\|$ is the distance of the point $\mathbf{b}$ from the line along $\mathbf{a}$. This distance is shortest when, $\mathbf{e} \perp \mathbf{a}$.

$$\mathbf{a}^\top(\mathbf{b}-\mathbf{p}) = \mathbf{a}^\top(\mathbf{b}-\alpha\mathbf{a}) = \mathbf{a}^\top\mathbf{b} - \alpha\mathbf{a}^\top\mathbf{a} = 0$$

$$\alpha = \frac{\mathbf{a}^\top\mathbf{b}}{\mathbf{a}^\top\mathbf{a}} \implies \mathbf{p} = \frac{\mathbf{a}^\top\mathbf{b}}{\mathbf{a}^\top\mathbf{a}}\mathbf{a}$$

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Orthogonal Projection onto Subspaces

- We can project vectors onto high dimensional subspaces.
- Consider the subspace $\mathcal{S} \subseteq \mathbb{R}^n$ spanned by the orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \cdots \mathbf{u}_r\}$.
- We want to project a vector $\mathbf{b} \in \mathbb{R}^n$ onto $\mathcal{S}$
  $\mathbf{b}_{\mathcal{S}}$ – the orthogonal projection of $\mathbf{b}$ onto $\mathcal{S}$ is given by the following,

$$\mathbf{b}_{\mathcal{S}} = \mathbf{U}\mathbf{U}^\top \mathbf{b}; \;\; \mathbf{U} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_r \end{bmatrix}$$

$$\text{Projection matrix } \mathbf{P}_{\mathcal{S}} = \mathbf{U}\mathbf{U}^\top$$

- A projection matrix is **idempotent**, i.e. $\mathbf{P}^2 = \mathbf{P}$.
- What does this mean in terms of projecting a vector on to a subspace?

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- A projection matrix is **idempotent**, i.e. $\mathbf{P}^2 = \mathbf{P}$.
- What does this mean in terms of projecting a vector on to a subspace? Projecting a vector that is already in the subspace results in the same vector.

Orthogonal Projection onto Subspaces

- Consider two matrices $\mathbf{U}_1, \mathbf{U}_2$ whose columns form an orthonormal basis of the subspace $\mathcal{S} \subseteq \mathbb{R}^m$, $\mathcal{C}(\mathbf{U}_1) = \mathcal{C}(\mathbf{U}_2)$.

- The projection matrix onto the subspace $\mathcal{S}$, $\mathbf{U}_1\mathbf{U}_1^\top = \mathbf{U}_2\mathbf{U}_2^\top$. We get the same projection matrix irrespective of which orthonormal basis one uses!

- Once we have the projection matrix $\mathbf{P}_{\mathcal{U}}$ onto a subspace $\mathcal{U}$, we get the orthogonal projection matrix onto $\mathcal{U}^\perp$ – the orthogonal complement of $\mathcal{U}$ – for free.

$$\mathbf{P}_{\mathcal{U}^\perp} = \mathbf{I}_n - \mathbf{P}_{\mathcal{U}}$$

Gram-Schmidt Orthogonalization

- Given a linearly independent set of vectors $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2, \cdots \mathbf{x}_n\}$, where $\mathbf{x}_i \in \mathbb{R}^m, \ \forall i \in \{1, 2, \cdots n\}$, how can we find a orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \cdots \mathbf{u}_n\}$ for $span\,(\mathcal{B})$? $\longrightarrow$ **Gram-Schmidt Algorithm**
- Its an iterative procedure that can also detect if a given set $\mathcal{B}$ is linearly dependent.

Data: $\{\mathbf{x}_i\}_{i=1}^n$
Result: Return an orthonormal basis $\{\mathbf{u}_i\}_{i=1}^n$ if the set $\mathcal{B}$ is linearly independent, else return nothing.
for $i = 1, 2, \cdots n$ **do**

1. $\tilde{\mathbf{q}}_i = \mathbf{x}_i - \sum_{j=1}^{i-1} \left(\mathbf{u}_j^\top \mathbf{x}_i\right) \mathbf{u}_i \longrightarrow$(**Orthogonalization step**);
2. **If** $\tilde{\mathbf{q}}_i = 0$ **then return**;
3. $\mathbf{u}_i = \tilde{\mathbf{q}}_i / \|\tilde{\mathbf{q}}_i\| \longrightarrow$(**Normalization step**);

end
return $\{\mathbf{u}_i\}_{i=1}^n$;

Gram-Schmidt Orthogonalization

- The algorithm can also be conveniently represented in a matrix form.

$$\mathcal{B} = \{\mathbf{a}_1, \mathbf{a}_2, \cdots \mathbf{a}_n\}$$

$$\text{Let } \mathbf{U}_1 = 0_{m\times 1} \quad \text{and} \quad \mathbf{U}_i = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_{i-1} \end{bmatrix} \in \mathbb{R}^{m\times(i-1)}$$

$$\mathbf{U}_i^\top \mathbf{x}_i = \begin{bmatrix} \mathbf{u}_1^\top \mathbf{x}_i \\ \mathbf{u}_2^\top \mathbf{x}_i \\ \vdots \\ \mathbf{u}_{i-1}^\top \mathbf{x}_i \end{bmatrix} \quad \text{and} \quad \mathbf{U}_i\mathbf{U}_i^\top \mathbf{x}_i = \sum_{j=1}^{i-1} \left(\mathbf{u}_j^\top \mathbf{x}_i\right) \mathbf{u}_j$$

$$\mathbf{u}_i = \frac{\left(\mathbf{I} - \mathbf{U}_i\mathbf{U}_i^\top\right)\mathbf{x}_i}{\left\|\left(\mathbf{I} - \mathbf{U}_i\mathbf{U}_i^\top\right)\mathbf{x}_i\right\|}$$

QR Decomposition

- Gram-Schmidt procedure leads us to another form of matrix decomposition – **QR decomposition**.

- Given a matrix $\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \in \mathbb{R}^{n \times n}$, whose columns form a linearly independent set.
  Gramm-Schmidt algorithm produces a orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \cdots \mathbf{q}_n\}$ for $\mathcal{C}(\mathbf{A})$.

$$\mathbf{q}_1 = \frac{\mathbf{a}_1}{r_1} \quad \text{and} \quad \mathbf{q}_i = \frac{\mathbf{a}_i - \sum_{j=1}^{i-1} (\mathbf{q}_j^\top \mathbf{a}_i) \mathbf{q}_j}{r_k}$$

where, $r_1 = \|\mathbf{a}_1\|$ and $r_k = \left\| \mathbf{a}_i - \sum_{j=1}^{i-1} (\mathbf{q}_j^\top \mathbf{a}_i) \mathbf{q}_j \right\|$.

QR Decomposition

$$\mathbf{a}_1 = r_1\mathbf{q}_1 \quad \text{and} \quad \mathbf{a}_i = r_i\mathbf{q}_i + \sum_{j=1}^{i-1}\left(\mathbf{q}_j^\top \mathbf{a}_i\right)\mathbf{q}_j$$

$$\begin{aligned}
\mathbf{A} &= \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n\end{bmatrix} \\
&= \begin{bmatrix}\mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n\end{bmatrix}\begin{bmatrix} r_1 & \mathbf{q}_1^\top\mathbf{a}_2 & \cdots & \mathbf{q}_1^\top\mathbf{a}_n \\ 0 & r_2 & \cdots & \mathbf{q}_2^\top\mathbf{a}_n \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_n \end{bmatrix} \\
&= \mathbf{QR}
\end{aligned}$$